

Segment - 2

Response time (The Role of Performance)

- ⊗ Response time
- ⊗ Bandwidth
- ⊗ Relative performance
- ⊗ clock cycle
- ⊗ clock rate
- ⊗ CPU execution time for a program
- ⊗ Measuring performance
- ⊗ CPU Time
- ⊗ CPI
- ⊗ MIPS

Response time / Execution time: The total time required for computation to complete a task including disk accesses, memory accesses, I/O activities, OS overhead, CPU execution time and so on.

The time between start and completion of a task

Throughput / Bandwidth: The number of tasks completed per unit time. The total amount of work done in a given time.

Relative performance: In discussing a computer design, we often want to relate the performance of two different computers quantitatively.

It is the ratio between two computer (suppose x and y)
Performance

$$\text{Performance}_x = \frac{1}{\text{execution time}_x}$$

- X is n times faster than Y it means

$$\frac{\text{Performance}_x}{\text{Performance}_y} = n$$

or,
$$\frac{\text{Performance}_x}{\text{Performance}_y} = \frac{\text{Execution time}_y}{\text{Execution time}_x} = n$$

Clock cycle: This is the time for one clock period, usually of the processor clock, which runs at a constant rate. It is also called as tick, clock period, clock or cycle.

Clock rate: Inverse of clock period.

It is the measurement of clock cycles per second.

$$\text{Clock rate} = \frac{1}{\text{clock cycle}}$$

Example: A computer having the clock rate of 4 GHz which has $\frac{1}{4 \text{ GHz}}$ of clock cycle.

CPU execution time for a program:

$$= \frac{\text{CPU clock cycles for the program}}{\text{clock rate}}$$

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Factors that affect the performance of computer:

- ① CPU clock rate
- ② CPI
- ③ Size of RAM
- ④ Speed of HDD
- ⑤ HDD space
- ⑥ GPU
- ⑦ Defragmenting files
- ⑧ Type of OS
- ⑨ Startup programs
- ⑩ Bad Registry entries

System performance:

Considering elapsed time on an unloaded system

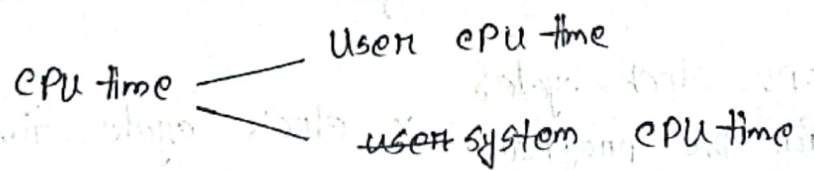
CPU performance: Considering user CPU time.

Measuring performance: This is the measure of computer performance.

CPU execution time / CPU time:

CPU execution time is the time the CPU spends computing for a task and does not include time spent waiting for I/O or running other program.

→ The actual time the CPU spends computing for a specific task. It is also called CPU time.



User CPU time - The CPU time spent in a program itself.

System CPU time - The CPU time spent in the OS performing tasks on behalf of the program.

Example: Unix time command: 90.7U 12.9S 2:39, calculate the percentage according to the total elapsed time.

Given that, unix time command: 90.7U 12.9S 2:39

U means user CPU
S means system CPU

90.7 → ~~90.7~~ 90.7 Sec

12.9 → 12.9 sec

2:39 → $(2 \times 60) + 39 = 159$

$$\therefore \frac{90.7 + 12.9}{159} = 0.65 \\ = 65\%$$

Ans: 9.7U 12.9S 2:39 65%.

CPI: Elaborated as clock cycles per instruction

→ It is the measurement of average number of clock cycles per instruction for a program or program fragment

$$\boxed{\text{CPI}} = \frac{\text{CPU clock cycles for a program}}{\text{Instruction for a program}} \times \text{Average clock cycles per instructions}$$

$$\text{CPU time} = \text{Instruction count} \times \text{CPI} \times \text{clock cycle Time}$$

$$= \frac{\text{Instruction count} \times \text{CPI}}{\text{clock Rate}}$$

Our favorite program runs in 10 seconds on computer A, which has a 400 MHz clock. We are trying to help a computer designer build machine B that will run this program in 6 seconds. The designer has determined that a substantial increase in the clock rate is possible, but this increase will affect the rest of the CPU design, causing machine A to run this program. What clock should we tell the designer to target?

Given that, CPU time_A = 10 seconds

$$\text{clock rate}_A = 400 \times 10^6 \text{ cycles/sec}$$

$$\text{CPU time}_A = \frac{\text{CPU clock cycles}_A}{\text{clock rate}_A}$$

$$\Rightarrow 10 = \frac{\text{CPU clock cycles}_A}{400 \times 10^6 \text{ cycles/sec}}$$

$$\Rightarrow \text{CPU clock cycles}_A = 10 \times 400 \times 10^6 \text{ cycles/sec}$$

$$= 4000 \times 10^6 \text{ cycles/sec}$$

$$\therefore \text{CPU time}_B = \frac{\text{CPU clock cycles}_B}{\text{clock rate}_B}$$

$$\Rightarrow \text{CPU time}_B = \frac{1.2 \times \text{CPU clock cycles}_A}{\text{clock rate}_B}$$

$$\Rightarrow 6 = \frac{1.2 \times 4000 \times 10^6 \text{ cycles}}{\text{clock rate}_B}$$

$$\Rightarrow \text{clock rate}_B = \frac{1.2 \times 4000 \times 10^6 \text{ cycles}}{6 \text{ seconds}}$$

$$= 800 \text{ MHz}$$

$\therefore$ Machine B must therefore have twice the clock rate of A to run the program in 6 seconds.

Q Suppose we have two implementations of the same instruction set architecture. Computer 1 has a CPI of 3.0 and a clock cycle time of 300 ps and computer 2 has a CPI of 4.0 and clock cycle time of 150 ps for same program, which computer is faster and by how much.

Solution: Given that, $CPI_1 = 3.0$

$$CPI_2 = 4.0$$

$$CC\ time_1 = 300\ ps$$

$$CC\ time_2 = 150\ ps$$

we know, $cpu\ time = \text{Instruction count} \times CPI \times \text{clock cycle time}$

Let I be the instruction count,

$$\text{so, } cpu\ time_1 = I \times 3.0 \times 300\ ps = 900I\ ps$$

$$\& \text{ } cpu\ time_2 = I \times 4.0 \times 150\ ps = 600I\ ps$$

$$\frac{Performance_2}{Performance_1} = \frac{CPU\ time_1}{CPU\ time_2} = \frac{900\ Ips}{600\ Ips} = 1.5$$

Hence, computer 2 is 1.5 times as fast as computer 1.

□ The hardware designer supplied -

Instruction class — CPI for this class

A

1

B

2

C

3

For a particular high-level languages statement, the compiler writer is considering two code sequences that require the following instructions count.

Code sequence — Instruction counts

1

2

1

2

2

4

1

1

→ which code sequence execute the most instructions?

→ which will be faster?

→ what is the CPI for each sequence?

Solution: ① Sequence 1 executes $2+1+2 = 5$ instructions
 Sequence 2 executes $4+1+1 = 6$ instructions
 So, sequence 2 executes most instructions.

② CPU clock cycle $= \sum (CPI_i \times C_i)$
 CPU clock cycle₁ $= (2 \times 1) + (1 \times 2) + (2 \times 3) = 10$ cycles
 CPU clock cycle₂ $= (4 \times 1) + (1 \times 2) + (1 \times 3) = 9$ cycles
 So, code sequence 2 is faster.

③ $CPI_1 = \frac{10}{5} = 2$ $[CPI = \frac{\text{CPU clock cycles}}{\text{Instructions count}}]$
 $CPI_2 = \frac{9}{6} = 1.5$

④ A program has 50% Load instructions, 30% store instruction and 20% Jump instructions. They take 4, 3, 2 CPI cycles respectively. Find average CPI.

Solution: CPI for load $= \frac{4}{\frac{50}{100}} = 8$

CPI for store $= \frac{3}{\frac{30}{100}} = 10$

CPI for Jump $= \frac{2}{\frac{20}{100}} = 10$

So, average CPI $= \frac{8+10+10}{3}$
 $= 9.33$

(Ans)

Million instruction per second

MIPS: It is a measurement of program execution speed based on the number of million of instructions

$$\begin{aligned} \text{MIPS} &= \frac{\text{Instruction count}}{\text{Execution time} \times 10^6} \\ &= \frac{\text{Instruction count}}{\frac{\text{Instruction count} \times \text{CPI}}{\text{Clock rate}} \times 10^6} \\ &= \frac{\text{Clock rate}}{\text{CPI} \times 10^6} \end{aligned}$$

Which computer has the higher MIPS rating? which computer is faster?

Measurement	Computer A	Computer B
Instruction count	10 billion	8 billion
Clock rate	4 GHz	4 GHz
CPI	1.0	1.1

Solution:

$$\begin{aligned} \text{MIPS}_A &= \frac{\text{Clock rate}_A}{\text{CPI}_A \times 10^6} \\ &= \frac{4 \text{ GHz}}{1.0 \times 10^6} = \frac{4 \times 10^9}{1 \times 10^6} \\ &= 4000 \end{aligned}$$

$$\begin{aligned} \text{MIPS}_B &= \frac{4 \text{ GHz}}{1.1 \times 10^6} \\ &= \frac{4 \times 10^9}{1.1 \times 10^6} \\ &= 3636.364 \end{aligned}$$

∴ Computer A has the higher MIPS rating.

Again, we know that, $MIPS = \frac{\text{Instruction count}}{\text{Execution time} \times 10^6}$

So, $\text{Execution time}_A = \frac{\text{Instruction Count}_A}{MIPS_A \times 10^6}$

$$= \frac{10 \text{ billion}}{4000 \times 10^6}$$

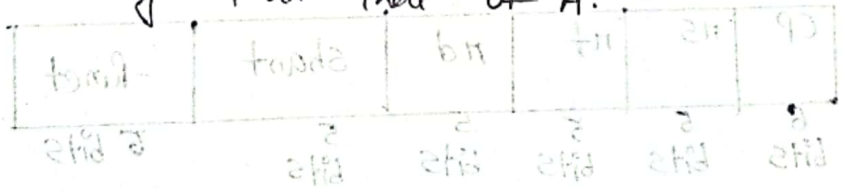
$$= \frac{10 \times 10^9}{4000 \times 10^6}$$

$$\text{And execution time}_B = \frac{8 \text{ billion}}{4000 \times 10^6}$$

$$= \frac{8 \times 10^9}{4000 \times 10^6}$$

$$= 2.25$$

Therefore, computer B is fast though computer B has a lower MIPS rating than that of A.



The basic operations of the instruction set are called instructions. The operation and the operand of an instruction are called the instruction fields.

The first register source operand is the first register source operand. The second register source operand is the second register source operand. The third register destination operand is the third register destination operand. The fourth register destination operand is the fourth register destination operand.

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